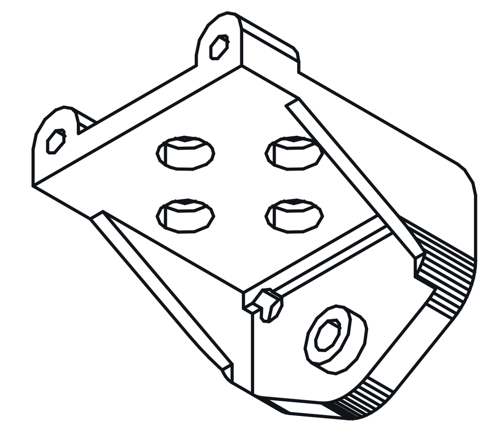
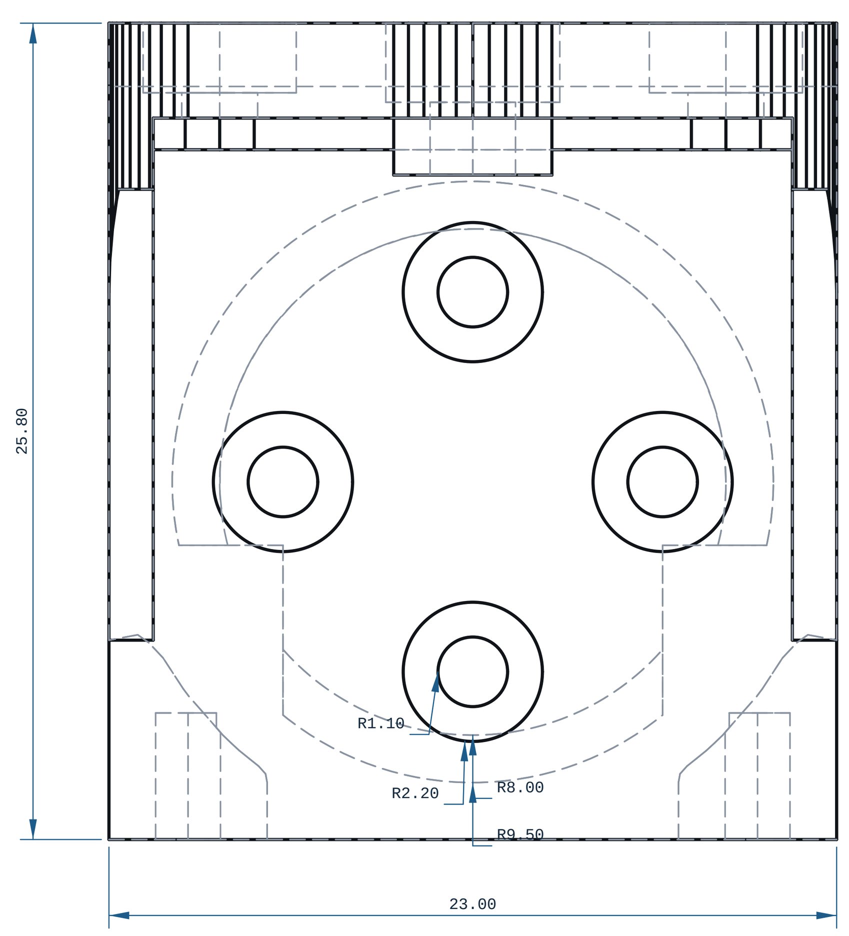
microduck-yaw2roll   SCALE 5:1   A1   THIRD ANGLE



ISOMETRIC VIEW

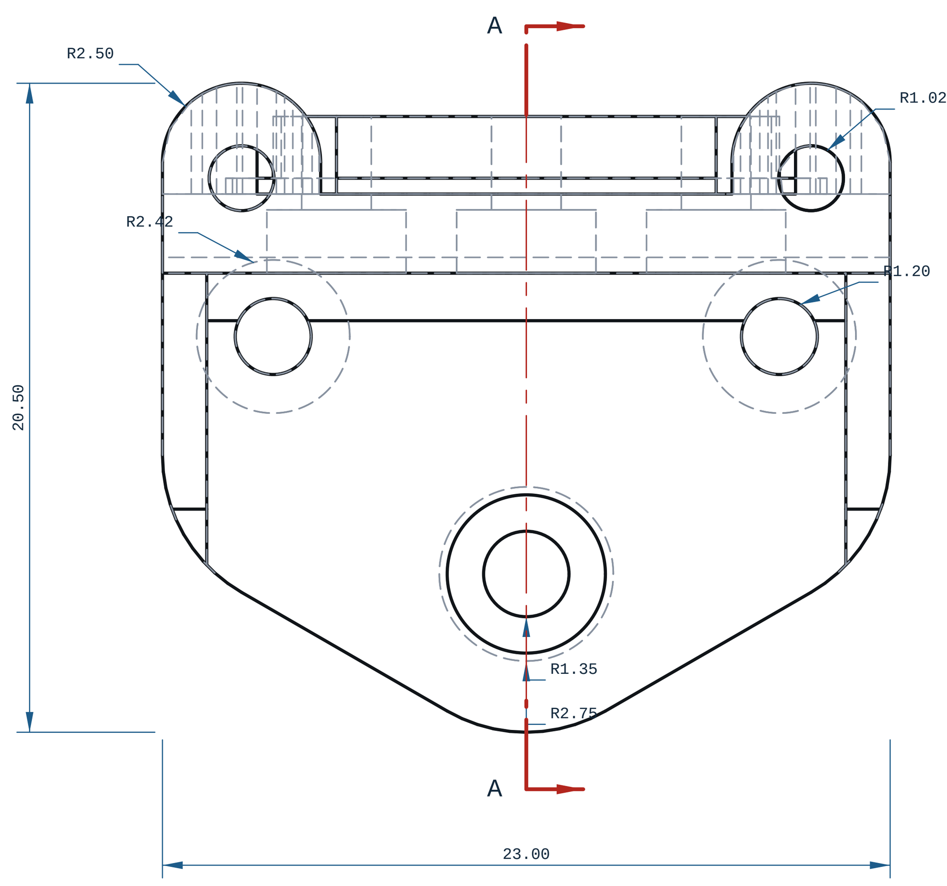
23.00 x 25.80 x 20.50 mm

REFERENCE RENDER (ISO2) ? rendered off this solid, 900 x 680 px



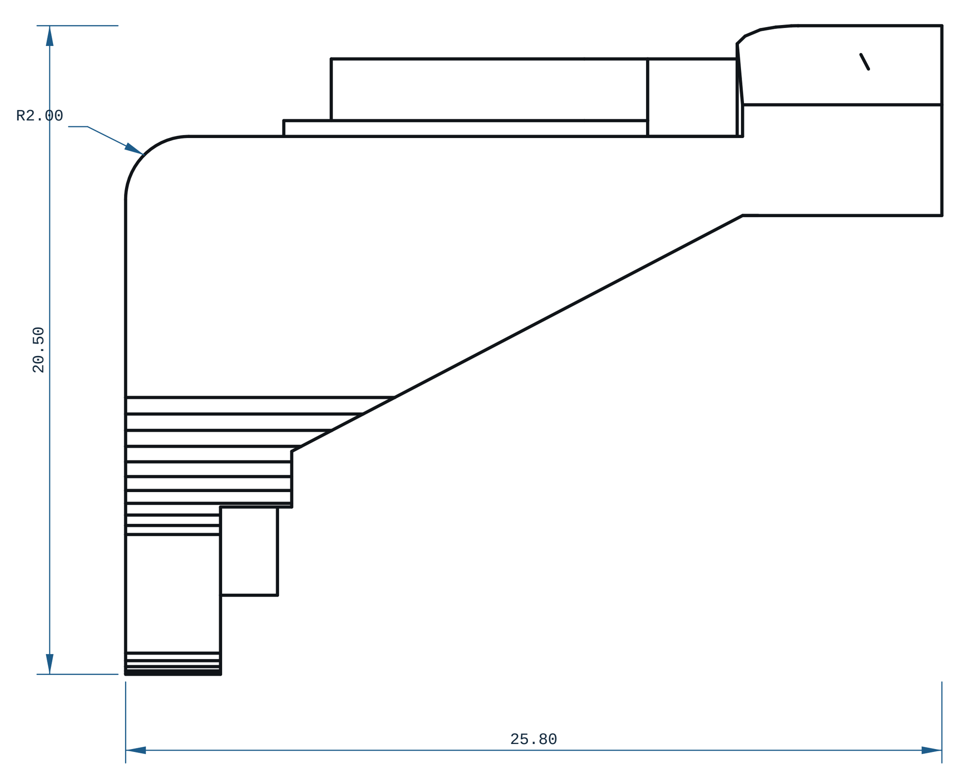
TOP VIEW

X ?   Y ?



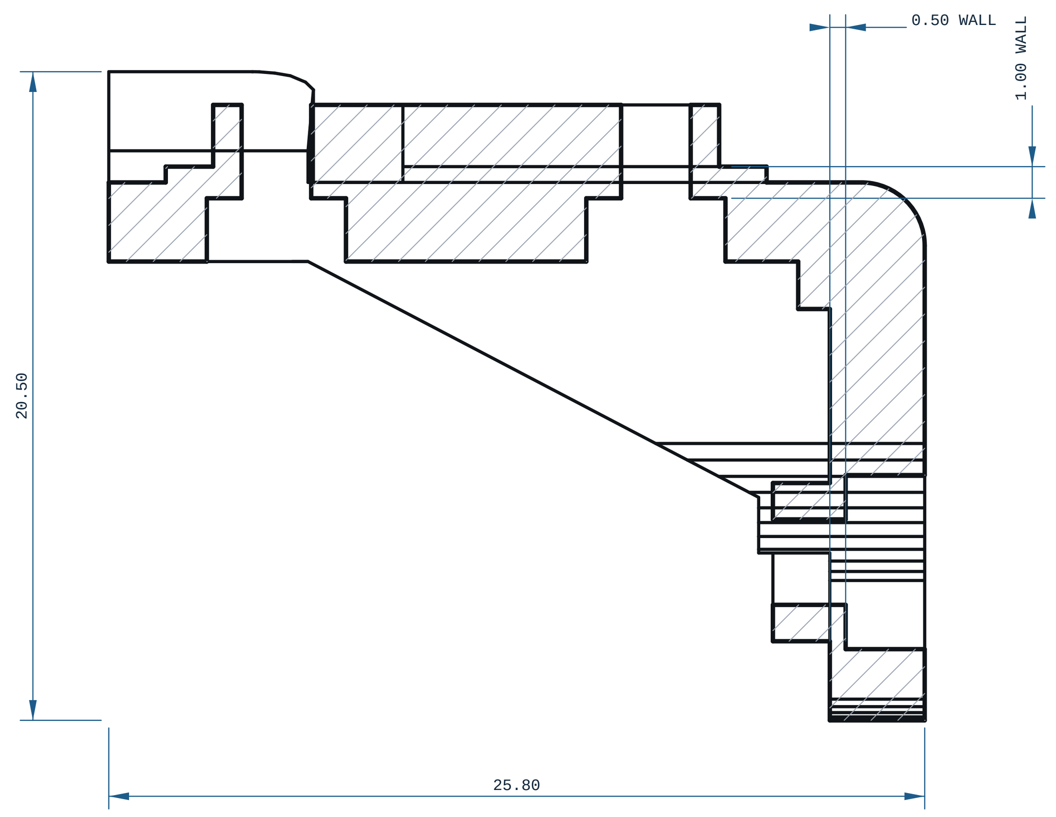
FRONT VIEW

-X ?   Z ?



RIGHT VIEW

-Y ?   Z ?



SECTION A-A

NOTES

1. ALL DIMENSIONS IN mm AND DEGREES. EVERY DIMENSION AND EVERY HOLE CALLOUT WAS MEASURED OFF THE SOLID.
2. PROCESS: FDM.
3. THINNEST WALL, WHOLE SOLID (RAY MINIMUM): 0.1980 mm, AT X=8.815 Y=-9.300 Z=12.999 ? MEASURED OVER THE WHOLE SOLID BY RAY+EXACT AT 0.9830 mm SAMPLE SPACING (A FEATURE NARROWER THAN THAT CAN BE MISSED). THIS IS A DISTANCE THROUGH MATERIAL, NOT A WALL: AT AN EDGE, A FILLET RUN-OUT OR A CHAMFER TIP IT IS REAL AND UNPRINTABLE BY DEFINITION. DO NOT JUDGE PRINTABILITY BY IT ? THE NEXT NOTE IS THE ONE THAT DOES.
4. PRINTABLE MINIMUM WALL: 1.1496 mm, AT X=18.062 Y=7.700 Z=2.040 ? MEASURED AS THE SMALLEST MEDIAN WALL OVER A NEIGHBOURHOOD OF 2.9500 mm (>= 8 SAMPLES), RAYS CAPPED AT 4.000 mm, BY cecad.inspect.printable_wall_detail. AGAINST THE 0.8000 mm FLOOR (2 PERIMETERS @ 0.4 mm NOZZLE, cecad/printed.py:801): VERDICT PASS
5. DETAIL CALLOUTS (MANUFACTURING-REQUIREMENTS A.7): NONE ON THIS SHEET, AND NONE WAS PROPOSED ? cecad.autosheet.detail_regions FOUND NO CLUSTER OF SMALL FEATURES DENSE ENOUGH TO BE WORTH ENLARGING ON THIS SOLID. EVERY FEATURE IS DIMENSIONED IN THE ORTHOGRAPHIC VIEWS AND THE HOLE TABLE.
6. GENERAL TOLERANCE CANNOT DETERMINE ? SEE PRINT / DFM NOTES UNLESS STATED.
7. SECTION A-A AT X=17.50: THINNEST MATERIAL IN THIS PLANE ONLY 0.50 mm, AT Y=9.75, Z=-2.73 ? A SCAN OF ONE CUT, NOT THE PART MINIMUM.
8. FRONT VIEW: hidden lines KEPT: without them 14 of 539 edge samples have no ink within 0.35 mm (worst 1.213 mm) and the feature would be missing from the sheet.
9. TOP VIEW: hidden lines KEPT: without them 80 of 490 edge samples have no ink within 0.35 mm (worst 3.862 mm) and the feature would be missing from the sheet.
10. RIGHT VIEW: hidden lines SUPPRESSED: all 406 edge samples still land on drawn ink without them.

PRINT / DFM

1. ORIENTATION: laid flat (23 x 26 x 20 mm)
2. THINNEST WALL, WHOLE SOLID (RAY MINIMUM): 0.1980 mm ? a distance through material, NOT a printable wall where it falls on an edge, a fillet run-out or a chamfer tip. Printability is judged on the next line.
3. PRINTABLE MINIMUM WALL: 1.1496 mm (smallest median wall over a 2.9500 mm neighbourhood, cecad.inspect.printable_wall_detail) against a 0.8000 mm floor (2 perimeters @ 0.4 mm nozzle): VERDICT PASS
4. 8 horizontal bore(s) bridge at the crown: ream any bearing or press-fit bore to size
5. FILAMENT: PLA, ~3 g (modelled), 103 layers @ 0.2 mm
6. TOLERANCE BASIS, IN-HOUSE: CANNOT DETERMINE. Every Bambu on this farm records tolerance_mm: null, cite "no coupon printed and measured on this machine" (ce-machines/bambu-h2s-*, bambu-h2d-*/capabilities.json, read 2026-09-02), and Bambu Lab's own H2D Pro spec sheet carries no dimensional-accuracy line. DIMENSIONS HERE ARE NOMINAL AS MODELLED.
7. TOLERANCE BASIS, OUTSOURCED: JLC3DP ±0.3 mm FDM (jlc3dp.com, ce-machines/farm-jlc3dp); Hubs ±0.5%, floor ±0.5 mm (hubs.com/3d-printing/, ce-machines/farm-hubs). Apply the route's figure, not both. The ±0.2 mm in cecad/machining.py fdm is a repo planning constant with no vendor cite and is NOT used here. One printed and measured coupon closes this note.
8. NO ISO 286 CLASS IS DECLARED ON ANY BORE: FDM cannot be sized to one (H7/p6 at Ø3 is a 0.010 mm band). REAM bearing and press-fit bores after printing.
9. ORTHOGRAPHIC VIEWS: 3 (FRONT, TOP, RIGHT), CHOSEN OFF THE BOUNDING BOX 23.0000 x 25.8000 x 20.5000 mm BY cecad.autosheet.choose_views ? THINNEST/LONGEST EXTENT RATIO 0.7946. THE FULL THIRD-ANGLE SET IS DRAWN.

MICRODUCK-YAW2ROLL

REV A

MATERIAL		PROCESS	SCALE
PLA		FDM	5:1
CATEGORY: bracket	GENERAL TOL: CANNOT DETERMINE ?~	FINISH: AS PRINTED	SHEET: A1 1/1
VOLUME (MEASURED): 2.83 cm3		MASS (ASSUMES PLA): 3.5 g	UNITS: mm / deg
SIZE (BBOX, MEASURED): 23.00 x 25.80 x 20.50		DRAWN: ce-cad 2026-09-03	PROJECTION: THIRD ANGLE
SOURCE (DESIGN FILE): ce-parts/microduck-yaw2roll/current/cad/part.py			